ISHAN P. BORKER

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Key Skills

Generative AI Technologies
RAG
LangChain
LLM
AWS Chatbot
Amazon Bedrock
BrowserBase
Perplexity.AI
Mosaic AI
Cline
Stagehand
Amazon Kendra
Amazon OpenSearch
Amazon Q Business
Amazon Q Plugin
Mesop
GenRocket
FastMCP
Agents
MCP
Amazon Q LLM
AgentSkills
Azure Functions
Terraform

Machine Learning
Python
FastAPI
Data Science
Supervised Learning
Unsupervised Learning
Computer Vision
NLP
Linear and Logistic Regression
Clustering algorithms
Decision Trees
Neural Networks
Reinforcement Learning
Feature engineering
Statistical analysis

Machine Learning libraries
Tensorflow
Keras
SpaCy
PyPDF
PyLDAVis
BERTopic
NLTK
PySpark
MLFlow

PROFILE

Accomplished software development professional with over 9.16 years of experience across diverse platforms, including Generative AI, Data Science, Machine Learning, NLP, Computer Vision, and Network Service Orchestration. Demonstrated expertise in Python, C++, FastAPI, MySQL, Tableau, Golang, Docker, Linux/Unix, POSTMAN, and JSON scripting. Proven track record in analyzing complex requirements, architecting robust solutions, and delivering end-to-end applications from design to deployment.
With 9.16 years of specialized experience in machine learning, I excel at studying client requirements, recommending optimal algorithms, performing advanced feature engineering, and creating insightful visualizations. My experience spans the entire project lifecycle, from pre-sales proposals and requirements gathering to system design, implementation, technical documentation, and customer engagement.
Highly adaptable and results-driven, I am dedicated to solving business challenges, embracing new technologies, and thriving in collaborative, fast-paced environments. I am eager to leverage my technical and management expertise to further my career in AI, contributing effectively to diverse and challenging portfolios.

EXPERIENCE

Persistent Systems, Verna, Goa as Engineering Lead, May 2022 - Present
Role: Jan 2026 – Present as **Engineering Lead**

- **UHG Optum Insights – Data: Dojo360 AI (Onsite Client: United Health Group):**
  [DX] E2E Testing of MCP Server post migration to the new Agentic Gateway: Executed end-to-end testing of Dojo360 MCP server following migration to the new Agentic Gateway, validating key MCP endpoints through Insomnia including listTools, listResources, listPrompts, listResourceTemplates, listmodules, searchprofile, searchmodule. Established secure connectivity between Dojo360 MCP Client and Agentic Gateway Production environment, generated and verified authentication tokens and successfully validated the list of Tools like ListDojo360 Modules, Search Dojo360 Documentation, Get Dojo360 Module Details, Dojo360 Module Profile Details, Dojo360 Module Changelog and Dojo360 Module Upgrade Guide through Insomnia
  [CMA] Setup Alarms in Splunk for Code Agent: Worked on Splunk On-Call alerting, workflow failure monitoring, severity-based notifications and configurable alert policies across CodeAgent. Added comprehensive unit tests for key failure scenarios including API errors, unauthorized access, service outages, Terraform timeouts and validation failures. Completed end-to-end validation using Insomnia and Splunk-On Call Dashboard
  DocSearch: Upgrade LLM to GPT-5: Upgraded DocSearch LLM deployment to GPT-5 and GPT-5.4-mini including configuration, code and deployment updates. Successfully validated /ai/search and /ai/generate APIs, completed testing in Insomnia and in dev environment in GitHub Actions and merged PR to the main branch
  Add CodeAgent API to https://api-dojo360.optum.com: Implemented CodeAgent API integration using FastAPI including API routing, configuration management, JWT Authentication, token validation and OpenAPI enhancements; Enabled and validated Swagger UI and OpenAPI endpoints for codeagent docs through Azure Function Apps; Developed authentication components like jwt auth, verify access token, added unit tests for JWT fallback and JWKS validation and ensured successful test execution; Configured Azure Front Door routing, URL reqrites and Terraform updates in dev environment; Updated OpenAPI security schemas, documentation handlers, environment configurations, dependencies and deployment artifacts; Extended similar API, Swagger, Authentication and Front Door integrations to dojo360-ai-docs-search service and updated front door terraform code in platform-infrastructure route mappings; Addressed PR feedback, supported EPL-CI workflow execution, deployed front door changes to dev environment
  [DX] Implement and publish new customer-facing Dojo360 skills: Developed and published customer-facing Dojo360 onboarding skills by creating skill manifests, prompts, instructions, smoke tests enabling streamlined user onboarding; Built deployment automation with deployment/rollback scripts, pipeline creation runbooks; Enhanced documentation, operational guidance and local execution support through comprehensive README updates and runbook creation; Designed and implemented HCPCTL-based pipeline bootstrap solution for automating ephemeral account onboarding and pipeline creation workflows; Collaborated with stakeholders to address PR feedback, validate solutions and drive publishing readiness
  [NxG] Validate NextGen compatibility in AWS ephemeral accounts: Validated NextGen compatibility in AWS ephemeral account by provisioning and verifying VPCs, Subnets and Security Groups using Terraform, AWS Portal and HCP Console; Automated AWS infrastructure deployment through Terraform and GitHub

Algorithms Classification Regression Clustering Time series forecasting Yolov3 Yolov4 OpenCV Mask R-CNN Faster R-CNN Named Entity Recognition Relationship Extraction Topic Modelling Latent Dirichlet Allocation Fasttext **Other Skills** Actionable Web Technologies JSON HTML JS PHP Curl commands Unix/Linux commands Golang Terraform YAML **OS** Linux/Unix Windows macOS **Database/BI tools** SQL MySQL Amazon S3 SpagoBI Tableau PowerBI Navicat GNU-Plot Amazon Aurora RDS MongoDB Compass **Python Libraries** NumPy Pandas SciPy Seaborn Scikit-learn Plotly-dash Matplotlib **Tools/Environments**	Actions, updating and executing terraform-plan.yml and infrastructure.yml workflows; Enhanced Terraform configurations for AWS networking and bootstrap automation; Implemented secure credential management using vaults, secrets and bootstrap modules supporting automated infrastructure provisioning; raised PR, validated acceptance criteria, delivered demo to stakeholders, drove process-improvement discussions and published comprehensive technical documentation [NxG] Validate NextGen compatibility in Azure ephemeral accounts: Validated NextGen compatibility in Azure ephemeral account by provisioning and verifying VNets and Subnets using Terraform and Azure Portal, automated Azure infrastructure deployment through Terraform and GitHub Actions by updating and executing terraform-plan.yml and infrastructure.yml workflows; Enhanced infrastructure modules and Terraform configurations for Azure networking and subscription bootstrap; Implemented secure cloud deployment practices using Azure Key Vault, secret management and GitHub repository integrations; raised PRs, validated acceptance criteria, resolved deployment issues and published technical documentation [DX] Compare MCP with Skills and provide recommendations: Designed and implemented a Terraform evaluation framework to compare MCP and Agent Skills approaches by enhancing agent metadata, tools and execution workflows; Established a structured evaluation pipeline (init -> plan -> apply -> destroy) with artifactory capture (logs, state, plan outputs) and standardized failure tracking ; Developed reusable prompts, instructions and skill templates with consistent YAML frontmatter to streamline agent, instruction and prompt creation; Covered automated evidence collection, failure documentation, comparison and reporting of infrastructure-as-code execution quality; Completed communication of skills created using MCP server and prepared detailed documentation covering points: comparison of Terraform Code Generation using MCP vs AgentSkills, guidance on when to use MCP vs AgentSkills [Skills] Develop Mechanism to update skills seamlessly: Designed and implemented a modular Agent Skills Framework aligned with the Agent Skills open standard to enable seamless skill updates without service restarts; built a dynamic skills loader for auto-discovery and execution, implemented MCP + Terraform setup, evaluated developer experience (auto-completion, error handling, performance), and documented framework, validation pipelines, versioning controls, CI/CD integration, centralized skills catalog (skills_catalog.yml) with metadata/governance, fallback handling, VS Code onboarding, testing suites, role-based access control, centralized hosting, monitoring, and performance tracking to ensure scalable, maintainable skill lifecycle management; Developed a standardized Skills Management Framework and centralized skills catalog (skills_catalog.yml) with structured metadata, governance, dynamic loading, filtering (tags, team, status) and fallback handling; implemented unified execution, validation pipelines, versioning controls, CI/CD quality checks and registry sync, improved developer experience via onboarding guides, VS Code integration and testing suites, enforced role-based access control, and established centralized hosting/monitoring for DX logs and performance to deliver enterprise-ready skill lifecycle management without agent restarts; Implemented the mechanism for automated skill updates and enhanced automation within the MCP server and prepared detailed documentation covering points: steps to set up Skill for MCP, reusability of skills to improve developer productivity, comparative summary and insights between MCP and Skills [TF Eval] [SPIKE] Validation Test Suite for Generated Terraform Code and Verify Full Stack Generation through TF Validation: Conducted research to evaluate a validation test suite for generated Terraform code, analyzing the Terraform Agent Test framework, proposing implementation strategies and authoring design documentation with technical recommendations; Implemented a comprehensive validation test suite for generated Terraform code using the Terraform Agent Test framework; designed and executed a multi-cloud (Azure, AWS, GCP) test bed covering multi-stack provisioning (App Service, Redis/Mem Cache, Cloudflare CDN), automated GitHub Actions workflows (terraform-plan, bootstrap), and end-to-end testing via ephemeral repositories and isolated branches—generated code through Insomnia, deployed across CSPs, validated terraform plan/apply for services (AWS ECS, Azure App Service/Function App), ensured correct tfstate and bootstrap execution, and tracked results in structured scoring sheets [MCP] Add workflows for MCP registry update: Implemented MCP registry publishing workflow by adding mcp-spec.json and publish-to-mcp-registry.yml to GitHub repository; updated ownership metadata and subdomain, successfully merged PR and executed the workflow to publish the MCP server [DX] Add tools and skill for Modules upgrade guide and changelog: Developed and integrated an MCP tool to generate module upgrade guides and changelogs using MongoDB-backed search, content parsing and JSON-RPC responses; implemented robust error handling and user prompts aligned with the standardized MCP dispatch pipeline; Refactored to use the shared Database Client (no direct MongoClient), made configurations fully config-driven and increased the unit test coverage to 80% based on PR reviews [DX] Add Dojo360 MCP to HCP API Gateway: Added AI MCP endpoint in platform API Gateway (YAML); updated front door in dojo360-platform-infrastructure Terraform (frontdoor local/prod/dev). Merged to main branch and deployed via GitHub. Collaborated with Agentic Gateway team to update MCP

EMS
Crosswork Workflow Manager (CWM)
Network Service Orchestration (NSO)
Lux
Jupyter Notebook
Google Colaboratory
Kaggle Notebook
Databricks Notebook
VS Code Jupyter Notebook
Amazon SageMaker Notebook
VSCode
PyCharm
Spyder
GitHub
Amazon EC2
Azure ML Studio
POSTMAN
Swagger UI
GitHub Copilot
Microsoft Copilot
AWS Code Whisperer
SonarQube
Docker
Insomnia
Microsoft Azure
CI/CD
GitHub Actions

Languages
English
Hindi
Konkani
Marathi
Sanskrit
Japanese (Basic)

Date of Birth
11/08/1993

Address
"ABHA", H.N.1794, Walkeshwar Wada, Betim, Bardez, Goa – 403101

Hobbies
- Playing indoor and outdoor games
- Gymming, Yoga
- Listening to music
- Reading books, newspapers and technical magazines
- Watching news and movies

server domain; added recommended tags to dojo360-ai-mcpserver-func-cus-dev resources in dojo360-platform-infrastructure; redeployed to dev, QA, and prod environments
CodeAgent & MCP Server Interaction: Created CodeAgent endpoint in dojo360-ai-code-agent repository with MCP client wrapper; tested payloads in Insomnia. Added unit tests (all passed) and end-to-end testing with negative/edge cases; presented client demo. Added MCP wrapper thin layer, error handling, logging, and retry logic to dojo360-ai-code-agent; created .md prompts for agentic profile selection (service names, Insomnia bodies, regex search from MongoDB without examples). Updated searcher/agent code to validate search API; completed negative/edge Insomnia tests
[Tool: Modules Docs] Develop tool to retrieve profile documentation and examples: Built tool to retrieve profile docs/examples: MCP docs tool queries CSP-specific modules_catalog in MongoDB for metadata. Added/tested API endpoints getProfileDocs/getProfileExamples in Insomnia for all modules/profiles per CSP. Identified correct profiles for validation; MCP client tested endpoints; MCP handled DB interactions via function app/tool decisioning. Tested filtering by module/profile from href; delivered client demo. Refactored code based on dojo360-ai-mcp-server changes using FastMCP; tested all API endpoints, added unit tests, improved coverage to 80%
MCP Client & Server Integration Testing: Conducted comprehensive integration testing for all MCP client API functions using Insomnia, covering unit tests, negative/edge cases. Verified end-to-end API endpoints; explored dojo360-ai-mcp-client-tool repository and MCP connection commands. Configured MongoDB connection; ingested docs/module catalog data into dojo360_docs database via Azure Blob Storage (JSON to embeddings)
Infrastructure & Documentation, Whitelist NextGen IP range for MCP Server: Collaborated with Enterprise Tech Infra to whitelist static/queue sites and NextGen MCP server IPs (bypass Azure), prepared and uploaded documentation to Dojo360 Wiki, activated Azure PIM, updated Function App ingress and integrated Azure Hub Commons Terraform; updated dynamic tags in tfvars across dojo360-ai repos, refreshed CI/CD workflows (CI.yml, CD.yml, create-release.yml, release-promotion.yml, copilot-setup-steps.yml) to ensure passing build checks and merged PRs to main, tested CMA JSON requests for AWS/Azure in Insomnia to validate profile picking and CSP mapping, fixed bootstrap and tfstate management workflows, reviewed generated Terraform code for accuracy, and resolved Dependabot/CodeQL vulnerabilities by updating dependencies and refining workflows
[Tool: Modules Catalog] Generate CSP Specific list of Modules, Profiles & Develop Modules Catalog Tool to Dojo360 MCP Server: Ingested modules catalog docs into MongoDB; performed local Insomnia testing (negative/edge cases) and raised PR. Built Browse Catalog tool for dojo360-ai-mcp-server: lists modules by technology (container/compute/database), exposes profiles, supports decisioning; backed by local MongoDB mock data. Implemented in VSCode Jupyter Notebook and server repository; reworked for CSPs (3 separate tools: one per CSP) using modular code (without Jupyter); updated CSV in modules_catalog collection. Added RAG pipeline unit tests; verified via Amazon Q LLM prompts ("list modules/profiles"); delivered client demo
Agentic MCP & Code Generator Enhancements: Developed Agentic MCP program accepting math expressions, using UAIS LLM for tool decisioning (add/subtract); built in VSCode Jupyter Notebook. Enhanced Code Generator Agent via Agentic Gateway: removed Tool Poisoning attack; tested APIs in Insomnia. Built MCP tool invoking LLM via FastMCP/Agents; compared responses in VSCode Jupyter Notebook
Dojo360-ai-feedback-api:Added and deployed the healthcheck endpoint, successfully validated via GitHub Actions and Insomnia; Deployed feedback api functionapp with managed identity for storage account;
Resolved security vulnerabilities for dojo360-ai-docs-ingestion and dojo360-ai-feedback-api, pushed the code changes in qa environment (Promotion Run CI/CD) and performed the validation

Role: Apr 2024 – Dec 2025 as **Lead Software Engineer**

- **UHG Optum Insights – Data: IBM CMOD (Onsite Client: United Health Group)**: Created evaluation templates for LLMs in Amazon Bedrock and documented AWS search service requirements, implemented lexical and semantic search in OpenSearch; handled claims data search and ingestion using Kendra and Q Business with S3 integration, compared search results from Kendra, Q Business, and OpenSearch; documented findings in tables and reports, explored Amazon Q plugin architecture in OpenSearch UI; set up Q-business index as a data source and provided accuracy feedback, configured Q-business index, demonstrated search scenarios, created custom plugins, and linked OpenSearch indices to S3 datasets; converted datasets to OpenAPI formats, assessed and compared search performance and latency across OpenSearch, Kendra, and Q Business using complex queries, prepared customer demo presentations, worked on image ingestion into OpenSearch by connecting S3 bucket of X-ray dataset to LLM, converting DICOM

• Watering plants	dataset into jpeg for ingestion, tested knowledge base of this dataset to Amazon Bedrock, developed front end UI for Aurora RDS database for patient records using mesop (insert and search operation), migrated the same code to Amazon SageMaker Notebook SOW1.0: Worked on Gen AI Tool and Milestone Mapping, explored the migration approach from DB2 to AuroraDB, and activities utilizing AWS Transform for acceleration, automation, and standardization, designed end-to-end architectural diagram for Query Pathway, evaluated and compared GenRocket versus custom Python approaches for large-scale, heterogeneous test data generation, documented requirements, explored generating and loading bulk data in GenRocket directly to Aurora, explored on the set up of GenRocket for storing EDI files in AWS S3 from an EC2 instance, configured GenRocket for bulk data generation and XML EDI output, and stored generated EDI data (single/bulk) in S3 from local system for business scenarios, exploring simultaneous data generation in S3 and uploading to AuroraDB using GenRocket • **Generative AI Platform development - Co engineering – Data (Client: Baldwin Risk Partners)**: Executed RAG classification and extraction for policy documents, explored chunking strategies for large, extracted tables via LLM calls. For the Integration of Amazon Bedrock Agent and AWS Chatbot with Microsoft Teams, modified SNS email testing to publish messages inside Teams Channel, created knowledge bases and Amazon Bedrock Agents, tested prompts, and developed an application within the Microsoft Teams developer portal for smooth integration. In Actionable Web Technologies, conducted a detailed tabular analysis of BrowserBase and Maxun focusing on scalability and browser dependencies; developed Python scripts using BrowserBase Playwright APIs to input URLs, download relevant PDFs, and enhanced stealth mode to bypass CAPTCHA and login processes. This was applied across several carrier websites such as CNA, Selective Flood, and Hartford, with metadata integration and LLM prompt-driven downloads using Stagehand approach. The Capabilities of Perplexity.AI were enhanced by improving prompt caching, dynamic updates, response accuracy, cache persistence, and query handling, with successful business feasibility testing conducted in Copilot UAT. Mosaic AI model was fine-tuned on the RAG Component Classification and Extraction dataset using Databricks, tested with GPT-35-turbo-SLM and meta-llama models, then deployed and thoroughly tested. Client requirements were addressed involving risk map automation, combining structured and unstructured data sources, implementing graph databases for RFP automation, utilizing LLMs with structured knowledge graphs, graph RAG, and knowledge-augmented graphs, along with preparing comprehensive documentation and detailed use case presentations for various proofs of concepts • **CISCO NSO Engineering (Client: CISCO):** Supported EMS adapter workflows (distribute, activate, commit image) on IOS-XE devices, enhanced API robustness by adding error handling and input validation in Go code, refactored EMS adapter for CW infrastructure compliance with authentication options, performed add image, activate and commit using install method for IOS-XE devices, executed EMS APIs via crosswork admin SVM using curl, updated documentation, created unit tests for refactored code, developed a Go client POC to fetch CCO catalog and SMU details, retrieved multiple devices in job summary EMS code **Role: May 2022 – Mar 2024** as **Senior Software Engineer** • **Routermation CISCO NSO Offshore Transition (Client: CISCO)**: Led Crosswork Workflow Manager (CWM) installation and development of adapters, resources, secrets, and EPNM; created and tested workflows and adapters for multiple applications using POSTMAN APIs (import, distribute, activate/commit image, job/device summaries, device upgrades, consolidated workflows). Implemented network device upgrades and adapters in Golang, executed workflows and SWIM APIs within the CWM UI. Managed Network Service Orchestration (NSO) installation and utilized Routermation Automation Kit for device onboarding, fleet/golden configuration upgrades. Performed LNT device upgrades via RAKfu Main workflow and lux test cases, resolved critical and major CW-EMS adapter issues using SonarQube, updated the CW-EMS adapter to support CWM 1.1 and integrated necessary libraries, and worked on IOS-XE device upgrade • **PDF Text Extraction and Code Testing (Client: IBM Infosphere)**: Extracted and preprocessed PDF text from URLs with PyPDF, generated histograms using NLTK, performed extractive summarization (LSA, Luhn, TextRank, LexRank), and measured inference times. Tested FastAPI code for FastText classifier on a text classification dataset (categories: Chemical Hazardous, Restricted Prohibited, Neither), recorded total training times. Computed volumetric stats for catalog files, including unique UNSPSC for Primary/Corrected Category, and data point availability and missing counts per UNSPSC for corrected category

- **Serving NER and REL models on clinical trials dataset using SpaCy:** Built NER and Relationship Extraction pipelines (tok2vec, various transformers), analyzed loss curves, enhanced performance metrics. Deployed English SpaCy and custom NER/REL models with FASTAPI, tested via POSTMAN and Swagger UI. Dockerized FASTAPI NER, deployed on AWS EC2, and pushed Docker container to GitHub
- **Topic Modelling**: Worked on topic extraction on docoh dataset of companies (Intuit, UHG, Wells Fargo, etc.) using packages like BERTopic, Latent Dirichlet Allocation (LDA), and Contextual Topic Modelling (CTM). Performed dimensionality reduction (PCA, UMAP), and clustering (HDBSCAN) techniques, leveraged transformers (RoBERTa, FinBERT, sentence transformers), visualized topics with PyLDAvis. Development is done using Python inside Google Colaboratory
- **Binary classification of network attacks**: Performed EDA, data preprocessing (matplotlib, seaborn), statistical analysis, feature engineering (one-hot encoding, label encoding, scaling, normalization), and feature selection (filter, wrapper, embedded methods). Conducted model selection with cross-validation and bootstrapping, trained multiple classifiers (Random Forest, Decision Tree, XGBoost, Gradient Boosting, Voting Classifier), optimized hyperparameters via GridSearchCV and RandomizedSearchCV, evaluated performance metrics, exported the trained model using joblib, and worked on model explainability using eli5, Partial Dependence Plots, ICE, LIME, and SHAP
- Developed MCQ part of the test using Python, created a dataset of question bank using .toml for AI Technothon event and development was done using Python, GitHub Copilot, and VS Code; worked on Medical Image segmentation using U-NET, Mask R-CNN on NVIDIA GPU Cloud (NGC)

TechnoPro India, Bengaluru as Machine Learning Engineer, Dec 2020 - Apr 2022

- **Annotation (Client: Nexar)**: Worked on semi-automated annotation using CVAT, ML algorithms like YOLOv3, Faster R-CNN, Mask R-CNN, and manual annotation using tools like Super Annotate, VoTT, LabelBox, etc. Led a team of annotators, and trained them on annotations like bounding box, polygon, polyline, cuboid, video, 3D annotation, instance, and semantic segmentation
- **Smart Farming**: Implemented tomato image classification using SVM, KNN, MLP, LR. Development is done using Python inside the Jupyter lab
- **Industry 4.0 (Client: Toyota Kirloskar Motors)**: Worked on detecting car parts in images and trained model using YOLOv3, tested code on car damaged parts detection using detectron2, Mask R-CNN, worked on predictive maintenance of bearing using regressors like SVR, RF, DT. Worked on Object Detection and Tracking of real-time intrusion using YOLOv4, OpenCV. Development is done using Python inside Google Colaboratory
- **Perimeter Intrusion Detection System (Client: Bengaluru International Airport Limited)**: Worked on POC, designing RFP on perimeter intrusion detection project for Airport surveillance covering aspects of Machine learning, Computer vision, RADAR for video surveillance, verification, tracking, access control security, and Smart Command and control system
- **Reviewer Analysis (Client: Nexar)**: Analyzed reviewer incident data in graphs using Python, matplotlib, and Seaborn, designed presentations, case studies, RFP, and POC for different AI-ML projects

Molecular Connections, Bengaluru as Senior Software Engineer, Jul 2020 – Oct 2020

- **EHR Predictive Modelling**: Cleaned EHR data, performed EDA, visualization, correlation, binary encoding on lab values, trained model, and exported model on the server using joblib, worked on FHIR Parser using Python, built HAPI FHIR Server by creating resources from JSON, tested REST operations on POSTMAN, created own FHIR Server

Rubixe, Bengaluru as Data Science Consultant, Dec 2019 - Jun 2020

- **Employee Performance Analysis**: Cleaned the data, performed EDA, created a visualization to find the department-wise performance of employees, and built a model to improve performance using ML algorithms like Random Forest, Gradient Boosting, XGBoost, ANN, KNN, LR, SVM, DT, used feature engineering, Grid & Randomized Search Cross Validation, SMOTE
- **Spare Parts Inventory Management**: Cleaned, analyzed the data, created the visualization, built a model using time series algorithms, and achieved better predictions for ARIMA
- **Improving ITSM**: Imported data from the server, performed EDA, implemented prediction on priority tickets, reassigned them using ML algorithms, used ARIMA, and SARIMA on incidents, achieved better predictions using Rolling Forecast
- **Sales Prediction**: Used SMOTE to handle imbalanced dataset, PCA for dimensionality reduction, predicted sales status using ML classification algorithms, plotted ROC, and precision-recall curve

- **Telecom Churn Prediction**: Predicted churn using ML classification algorithms, used K-fold, Stratified K-Folds Cross Validation to improve performance, plotted ROC, precision-recall curve, calculated churn-risk score, exported model on the server
- **Bank Credit Score**: Imported Customer Accounts, Demographics, and Enquiry data from the server, created a visualization to check factors influencing customers having a good or bad credit history, achieved 95% accuracy using ML classification algorithms

Digiapt Software, Bengaluru as Associate Software Developer, Oct 2019 - Nov 2019

- Categorized the data of companies using Tableau, plotting their locations using My Maps, used Snovio to capture emails of employees and analyze roles

Eaglys Inc. (Deputed from Teczuno), Tokyo, Japan as Research Engineer, Dec 2018 – Jul 2019

- **Back Office Manhour Management**: Generated reports using graphs, and presentations for attendance spreadsheet from Google Drive
- **Secure DB**: Tested SQL queries, achieved API documentation using Natural Docs, PyDoc

Teczuno Global India, Bengaluru as Data Analyst, Apr 2018 - Aug 2018

- Gathered data, developed an application to locate startups using Tableau, and built-up concepts as a part of pre-sales support

Anant Infomedia, Panaji, Goa as Project Trainee, Aug 2017 - Feb 2018

- **Employee Attrition**: Predicting employee attrition using classification, regression, clustering, and anomaly detection inside Azure ML Studio
- **Business Intelligence**: Used SpagoBI for data analysis, created dashboards, cockpits, reports and deployed on SpagoBI server, worked on PHP, JS, HTML, Navicat, MYSQL

INTERNSHIP EXPERIENCE

Rubixe, Bengaluru, Data Science Intern, Mar 2020 - May 2020

- **AI for Hiring**: Designed business proposal for student placement, built model using machine learning classification algorithms for placement prediction, and published paper in International Journal of Science and Research (IJSR)

Goa Electronics Limited, Goa, Project Intern, Oct 2018 – Nov 2018

- Designed Software Requirement Specification (SRS) for labor-employment, land dispute, and fire-emergency services, and created flowcharts for employment exchange, worked on the analysis of making compliant government websites

C-DAC ACTS, Pune, Project Intern, Jun 2016 – May 2017

- MTech Project Work: Implemented code of backpropagation using C in Ubuntu, used OpenMP technique to execute parallel code to improve efficiency, benchmarking done on Intel 64-bit, Intel Xeon architectures, achieved better performance on Intel Xeon Phi, carried out dissertation work and presented a paper in the International Workshop on Internet of Things and TV White Spaces (WIOT' 2017), published research paper in the International Journal of Current Research (IJCR)

CSIR-NIO, Goa, Project Intern, Dec 2013 – Jan 2014

- Research on robotic OS, implemented a serial port program to transfer data from one USB port to another in Ubuntu

EDUCATION

- M. Tech High Performance Computing at C-DAC and Veltech University, 2015 – 2017, 8.61 CGPA
- B.E. Computer Engineering at Goa College of Engineering (Goa University), 2011 –2015, 70%
- Higher Secondary School (Class XII) – Science, Goa Board, 2010-2011, 74%
- High School (Class X) – General, Goa Board, 2008-2009, 82.17%

TECHNICAL CERTIFICATIONS/ACHIEVEMENTS

- Neo4j Certification: Neo4j Fundamentals, Cypher Fundamentals, Graph Data Modeling Fundamentals, Using Neo4j with Python, Intermediate Cypher queries
- Anthropic Claude Certification: Claude 101, Introduction to Agent Skills, Claude Code 101, Claude Code in Action, Introduction to Claude Cowork, Introduction to Model Context Protocol, AI Fluency Framework & Foundations, AI Capabilities and Limitations, AI Fluency for educators, AI Fluency for students, Introduction to subagents, Teaching AI Fluency, AI Fluency for nonprofits, AI Fluency for Small Businesses, Model Context Protocol_ Advanced Topics, Building with the Claude API, Claude Platform 101, Claude with Amazon Bedrock, Claude with Google Cloud's Vertex AI
- Persistent Systems: High Five Individual Awards – Completed assignment on IBM-CMOD
- Persistent Systems: SASVA Immersive Workshop
- Databricks: Knowledge Badge - Get Started with Data Governance on Databricks, Building RAG Agents with Agent Bricks, Deploying and Monitoring Agent Applications on Databricks, Building Reliable Conversational Agents with Genie, Data Governance at Scale
- Databricks: Generative AI Application Evaluation and Governance, Get Started with Data Governance on Databricks and Get Started with Databricks for Data Engineering
- Databricks Delivery Specialization: CDW Migration Best practices, EDW Migration Best practices
- Databricks: Generative AI Application Evaluation and Governance, Generative AI Solution and Application Development, Generative AI Application Deployment and Monitoring
- Persistent Systems: CursorAI Course, GitHub Copilot Course
- Databricks: Knowledge Badge - Generative AI Application Development, EDW-ETL Migration to the Data Intelligence Platform
- Partner Training - Financial Services Industry Specialization for Gen AI & LLM Databricks Badge
- Partner Training - Healthcare & Life Sciences Industry Specialization for Gen AI & LLM Databricks Badge
- Snowflake: Snowflake Sales Professional Accreditation, Snowflake Beginner Fundamentals
- Persistent Systems: iAura Course, Generative AI Foundation Course
- LinkedIn: Preparing for the Future of Work with AI Agents, Preparing Your Organization for AI Agents, Leading through AI Agent Disruption, Transforming Business with AI Agents: Autonomous Efficiency and Decision-Making, Fundamentals of Agentic AI: Business Implications and Ethical Insights, Cybersecurity Foundations
- Persistent Systems: Participated in Semicolons 2026, Annual Global Hackathon of Persistent Systems
- Persistent Systems: Gen AI Competency Level - 3 Certification
- Persistent Systems: Participated in Semicolons 2025, Annual Global Hackathon of Persistent Systems
- Persistent Systems: Python ML Competency Level - 3 Certification, Python ML Competency Level - 2 Certification
- Persistent Systems: Generative AI for Practitioners Certification (Objective and Subjective), GenAI Foundation – GenAI Assisted Coding using Amazon Code Whisperer
- Databricks Accredited Generative AI Fundamentals
- Career Essentials in Generative AI by Microsoft and LinkedIn
- Databricks Lakehouse Machine Learning Associate
- Persistent Systems: Participated in Semicolons 2023, Annual Global Hackathon of Persistent Systems
- Fundamentals of the Databricks Lakehouse Platform Accreditation
- Persistent Systems: Information Security Awareness Secure Coding/Data Handling & Privacy Training
- IABAC, Amsterdam: Certified Data Scientist & Data Science Foundation by
- Deep Learning specialization by Coursera and deeplearning.ai on Coursera.org
- Certified Data Scientist course completion certificate by DataMites, Bengaluru
- Paper publication "Parallelization of Backpropagation Algorithm & Benchmarking" in IJCR
- Paper publication "Artificial Intelligence for Hiring" in IJSR
- Paper presentation on "Backpropagation Algorithm & Use of OpenMP in ML" in International Workshop on IOT & TV White Spaces
- Udemy: Agentic AI Playbook Complete Guide for Tech Leaders
- Udemy: Context Engineering Masterclass LLMs, RAG & Agents
- Udemy: Agentic Workflows Design End-to-End AI Processes
- Udemy: Master Class for Agentic AI Design Patterns
- Udemy: LangChain & LLMs - Build Autonomous AI Tools Masterclass
- Udemy: Prompt Engineering for Everyone Bootcamp

	• Udemy: Generative AI with Context: RAG, CAG & KAG Applications • Udemy: Agentic AI Production Grade AI Agents using CrewAI and AWS • Udemy: Codex - The Practical Guide • Udemy: Introduction to Generative AI and LLMs • Udemy: Foundations in Responsible AI & AI Ethics • QC101 Quantum Computing & Intro to Quantum Machine Learning • Udemy: Building AI Agents & Agentic AI System via Microsoft Autogen • Udemy: Ultimate RAG Bootcamp Using LangChain, LangGraph & LangSmith • Udemy: Building Gen AI App 12+ Hands-on Projects with Gemini Pro • Udemy: Complete MCP Bootcamp- Build Next-Gen AI Agents with MCP • Udemy: Full stack generative and Agentic AI with python • Udemy: MCP Masterclass - Complete Guide to MCP in Python [2025] • Udemy: Intro to AI Agents and Agentic AI • Udemy: LangGraph- Develop LLM powered AI agents with LangGraph • Udemy: Agentic AI for Beginner • Udemy: Terraform on Azure • Udemy: Terraform on AWS for Beginners Real-World Project • Udemy: Terraform for the Absolute Beginners with Labs • Udemy: 2026 Bootcamp Generative AI, LLM Apps, AI Agents, Cursor AI • Udemy: The Complete Prompt Engineering for AI Bootcamp (2025) • Udemy: Intro to MCP • Udemy: 2025 - Understand, Test, Fine-tune AI Model with HuggingFace • Udemy: Generative AI solutions with Azure OpenAI • Udemy: Learn Hugging Face Bootcamp • Udemy: Mastering Generative AI with PyTorch - Hands-on Experience • Udemy: Evaluating AI Agents • Udemy: Chatbot - The Development Guide 2025 (Beginner + Advanced) • Udemy: Basic to Advanced - Retrieval-Augmented Generation (RAG) • Udemy: Langchain for beginners - Build GenAI LLM Apps in Easy Steps • Udemy: LangGraph in Action - Develop Advanced AI Agents with LLMs • Udemy: Intro to Google's A2A Protocol - Interoperable AI Agents • Udemy: AI Agentic Design Patterns with Ollama & OpenAI Guide • Udemy: Amazon Q Developer for Programmers and DevOps AWS AI coding • Udemy: Responsible AI - Principles, Practices, and Applications • Udemy: Prompt Engineering for Beginners - Learn ChatGPT Prompting • Udemy: LLM Fine Tuning Fundamentals + Fine tune OpenAI GPT model • Udemy: LLMs Mastery - Complete Guide to Transformers & Generative AI • Udemy: Generative AI for Beginners • Udemy: Intermediate Python - Memory, Decorator, Async, Cython & more • Udemy: Learn Selenium with Python, PyTest & Frameworks • Udemy: Serverless using AWS Lambda for Python Developers • Udemy: AWS MasterClass - Monitoring and DevOps with AWS CloudWatch • Udemy: AWS Security Best Practices • Udemy: Introduction to Virtualization - 90 Minute Crash Course • Udemy: AWS EC2 Crash Course + Load Balancing with Demos and PDFs • Udemy: Learn Basic DB2 on Mainframe for Beginners • Udemy: DB2 For Beginners • Udemy: Containers on AWS ECS, EKS, and Fargate 2 hour Crash Course • Udemy: AWS Serverless Lambda Functions 2 hr. Hands On Crash Course • Udemy: Hands-On Introduction to CloudFormation - 2 hour course • Udemy: Amazon Aurora PostgreSQL from A to Z • Udemy: Git Going Fast One Hour Git Crash Course • Udemy: Complete Agentic AI Bootcamp with LangGraph and Langchain • Udemy: AWS DynamoDB 2025: High Performance Strategies Included • Udemy: Complete Generative AI Course With Langchain and Huggingface

- Udemy: AWS RDS & Aurora Crash Course – Zero to Hero in 3 Hours!
- Udemy: Master RAG: Ultimate Retrieval-Augmented Generation Course
- Udemy: AWS Certified Machine Learning Engineer Associate: Hands On
- Udemy: The Absolute Beginners Guide to Cyber Security 2025 - Part 1, 2, 3 and 4
- Udemy: Introduction to Cyber Security - 2 Hour Crash Course
- Udemy: AWS CDK for professionals (Python and TypeScript)
- Udemy: AWS SAM Framework and AWS Lambda - A Complete Hands-on Guide
- Udemy: AWS Lambda - A Practical Guide - Learn from an Expert
- Udemy: Learn AWS Identity Management with AWS IAM, SSO & Federation
- Udemy: AWS Identity Access Management (IAM) Practical Applications
- Udemy: Introduction to AWS - Understand AWS basics in 3 hours
- Udemy: Gemini Google AI: The All-in-One AI Masterclass 2025
- Udemy: Mathematics Behind Large Language Models and Transformers
- Udemy: Introduction to Large Language Models (LLMs) In Python
- Udemy: MCP Crash Course Complete Model Context Protocol in a Day
- Udemy: Intro to Large Language Models (LLMs)
- Udemy: Automate the Boring Stuff with Python Programming
- Udemy: Complete Python Bootcamp 2024: Zero to Expert in Python
- Udemy: AWS SageMaker Practical for Beginners Build 6 Projects
- Udemy: MongoDB – The Complete Developer's Guide 2024
- Udemy: Complete Tensorflow2 and Keras Deep Learning Bootcamp
- Udemy: Taming Big Data with Apache Spark and Python – Hands-on
- Udemy: Natural Language Processing with Python
- Udemy: Natural Language Processing with Transformers in Python
- Udemy: LangChain with Python Bootcamp
- Udemy: REST APIs with Flask and Python in 2024
- Udemy: Python for Data Science and Machine Learning Bootcamp
- Udemy: Complete Data Science, Machine Learning, DL, NLP Bootcamp
- Udemy: Git for Geeks: Quick Git Training for Developers
- Udemy: Amazon Bedrock & AWS Generative AI [Beginner to Advanced]
- Udemy: Generative AI: Beginner to Pro with OpenAI & Azure OpenAI
- Udemy: LangChain- Develop LLM-powered applications with LangChain
- Udemy: Deployment of Machine Learning Models
- 1st prize in National Summer Training Program on "Bigdata & Hadoop" by Revert Technology, EDC-IIT Roorkee
- Diploma Course in Java programming with JDBC and C programming at MICE

NON-TECHNICAL CERTIFICATIONS

- Udemy: Artificial Intelligence and Business Strategy AI Roadmap
- Udemy: Become a 10x Product Manager with ChatGPT & Generative AI
- Udemy: The EU AI Act Explained: Compliance and Practical Impact
- Udemy: Advanced Product Management – Vision, Strategy and Metrics
- Udemy: AI Ethics/ Responsible Use
- Udemy: AI Leader: Generative AI & Agentic AI for Leaders & Founders
- Udemy: Mastering Workplace Communication Skills
- Udemy: AI for Product Management & Innovation
- Udemy: Product Management & Product Design with Generative AI
- Udemy: Generative AI for Leaders: Strategy & Innovation
- Udemy: Project Management with Gen AI: A Guide for Project Managers
- Udemy: Business Communication Skills: Business Writing & Grammar
- Udemy: Generative AI for Leaders & Business Professionals
- Udemy: Entrepreneurship - How To Start A Business From An Idea
- Udemy: Generative AI for Business Leaders: A Quick Overview
- Udemy: Product Management 101
- Udemy: Project Management Fundamentals

	• Udemy: Leadership: Practical Leadership Skills • Udemy: Business Communication Skills: Handling Clients & Customers • Udemy: Communication Skills for Professionals • Udemy: Communication Skills: Become Clear, Concise, Confident • Udemy: Delegation Strategies for Managers • AMCAT certification exam - 77.72 percentile, Apr 2017 **DECLARATION** I hereby declare that all the information provided above is true to my knowledge. **Sign:** **Date:**